VELAMMAL
INSTITUTE OF TECHNOLOGY
Chennai - Kolkata Highway, Panchetti, Ponneri - 601 204


INSTITUTION'S INNOVATION COUNCIL (Ministry of HRD Initiative)

# DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

## ET3491-EMBEDDED SYSTEMS AND IOT DESIGN

## MINI PROJECT REPORT

## On

# FIRE FIGHTING ROBOT

## ACY:2025-2026(EVEN SEMESTER)

## Batch:2023-2027

## III/VI/A

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# CERTIFICATE

Certified that this project report titled "  FIRE FIGHTING ROBOT" is the bonafide work of Anusri K(113323106004),Danishtaa U(113323106018),Sridharshini V S(113323106097),Sharmila S(113323106094),Thiriloshani S(113323106108)and Vaishnavi S(113323106111)who carried out the mini project work under my supervision for the course ET3491-Embedded systems and IOT design during the academic year 2025-2026.

The work has been completed under the guidance of the faculty and is submitted to Velammal Institute  of Technology.

| | |
|---|---|
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# ABSTRACT

Fire accidents pose a serious threat to human life, property, and the environment, especially in industrial areas and confined spaces where human access is difficult. Early detection and immediate response are crucial to minimize damage and ensure safety. This project focuses on the design and development of an IoT-based firefighting robot using the ESP32 microcontroller to address this problem effectively.

The main objective of the project is to develop an automated system capable of detecting fire, extinguishing it, and sending real-time alerts to authorities. The system uses multiple flame sensors and an optional smoke sensor to identify fire presence accurately. Once fire is detected, the ESP32 activates a water pump through a relay module to suppress the fire. A servo motor helps in directing the water flow towards the fire source, and a motor driver enables the robot to move towards the affected area.

Additionally, the system sends instant notifications via WhatsApp using IoT technology, ensuring timely communication. This project provides a low-cost, efficient, and safe solution for firefighting applications.

# INTRODUCTION

Fire accidents are one of the major hazards that cause serious damage to human life, property, and the environment. They can occur in homes, industries, warehouses, and public places due to various reasons such as electrical faults, gas leaks, or human negligence. In many cases, fires spread rapidly and become uncontrollable before proper action can be taken. Moreover, certain fire-prone areas are difficult and dangerous for humans to access, making firefighting a risky task. Hence, there is a strong need for an efficient and automated system that can detect and control fire at an early stage.

With the advancement of technology, the Internet of Things (IoT) has emerged as a powerful tool in developing smart safety systems. IoT enables devices to communicate with each other through the internet and share real-time data. This technology plays a crucial role in monitoring environmental conditions and sending instant alerts during emergencies. In fire safety applications, IoT helps in providing timely notifications to authorities, thereby reducing response time and preventing major damage.

Traditional firefighting methods rely heavily on human intervention, which may not always be safe or effective, especially in hazardous environments such as chemical industries, oil refineries, and confined spaces. In such situations, the use of robotic systems can significantly reduce human risk. A firefighting robot can navigate towards the fire source, detect the presence of fire using sensors, and take immediate action to extinguish it.

The problem addressed in this project is the absence of a low-cost, automated firefighting system that can both detect and suppress fire while also providing real-time communication to concerned authorities. Many existing systems either focus only on detection or require manual control, which limits their effectiveness.

The main objective of this project is to design and develop a smart firefighting robot using the ESP32 microcontroller. The system is equipped with multiple flame sensors and an optional smoke sensor for accurate fire detection. Once fire is detected, a water pump is activated through a relay module to extinguish the fire, and a servo motor helps in directing the water flow. The robot is made mobile using a motor driver, enabling it to move towards the fire location. Additionally, the system uses IoT technology to send instant WhatsApp alerts to predefined authorities, ensuring quick response and improved safety.

This project aims to provide a cost-effective, reliable, and efficient solution for fire detection and control, reducing human involvement and enhancing overall safety in critical environments.

# RELEVANCE TO SUSTAINABLE DEVELOPMENT GOALS(SDG)

This project supports the following United Nations Sustainable Development Goals (SDGs): SDG 3 – Good Health and Well-being, SDG 9 – Industry, Innovation and Infrastructure, SDG 10 – Reduced Inequalities, and SDG 17 – Partnerships for the Goals.

SDG 3 – Good Health and Well-being:

The firefighting robot helps in protecting human life by detecting and extinguishing fire at an early stage. It reduces injuries, health risks, and fatalities caused by fire accidents, thereby promoting safety and well-being.

SDG 9 – Industry, Innovation and Infrastructure:

The project involves the use of modern technologies such as ESP32, sensors, and IoT communication. This promotes innovation and contributes to the development of smart and reliable safety infrastructure.

SDG 10 – Reduced Inequalities:

The system is designed to be low-cost and efficient, making it accessible to both rural and urban areas. This ensures that advanced safety solutions are available to a wider population, reducing inequality.

SDG 17 – Partnerships for the Goals:

The integration of WhatsApp alert system enables communication between users and authorities. This supports coordination and collaboration during emergency situations.

Thus, the project contributes to safety improvement and technological advancement by combining automation, IoT, and real-time communication.

# LITERATURE REVIEW

| S.NO | TITLE | INFERENCE | LIMITATIONS |
|---|---|---|---|
| 1 | Fire fighting robot using Arduino IEEE | Detects fire and estinguish using pump | No alert system |
| 2 | IOT based fire detection system | Detects fire and send alert | No estinguishing system |
| 3 | Autonomous fire fighting robot | Moves and estinguish fire automatically | High cost |
| 4 | Smart Fire alert system usingGSM | Send SMS alerts | No mobility |
| 5 | Wireless sensor network for fire montoring | Monitors fire using sensors | High cost and maintainence |

# EXISTING METHOD

Existing fire detection and firefighting systems mainly rely on manual operation or basic automated alert mechanisms. In traditional methods, fire is detected using standalone smoke detectors or flame sensors, which trigger alarms to alert people nearby. However, these systems only provide warning and require human intervention to control or extinguish the fire. This delay can lead to severe damage, especially in critical environments.

Some modern systems use GSM or IoT technology to send alerts through SMS or mobile applications. Although these systems improve communication, they still lack the capability to take immediate action to suppress the fire. In addition, most existing firefighting robots are either remotely controlled or designed with complex architectures, making them costly and less accessible for common use.

Another limitation of existing systems is the absence of mobility and real-time decision-making. Many fire detection systems are fixed in one location and cannot move towards the fire source. This reduces their effectiveness in large or dynamic environments where fire can spread quickly.

Overall, the existing methods focus mainly on detection and alerting, but they lack an integrated approach that combines fire detection, automatic extinguishing, mobility, and real-time communication. These limitations highlight the need for a more efficient, low-cost, and intelligent firefighting system.

# PROBLEM STATEMENT AND SOLUTION

## STATEMENT:

Fire accidents pose a serious threat to human life, property, and the environment. In many situations, fires occur in areas that are difficult or dangerous for humans to access, leading to delays in detection and response. Existing fire detection systems mainly provide alerts without taking immediate action to control the fire. These systems depend on human intervention, which increases the risk of injury and damage. Moreover, most systems lack mobility and real-time communication, making them less effective in dynamic and large-scale environments. Therefore, there is a need for an automated system that can detect fire, respond quickly, and reduce human involvement.

## SOLUTION:

To address this issue, an IoT-based firefighting robot using ESP32 is proposed. The system uses flame sensors and a  to detect fire. Once detected, a water pump is activated through a relay to extinguish the fire, and a servo motor directs the water flow. The robot is made mobile using a motor driver to reach the fire location.

Additionally, the system sends WhatsApp alerts to authorities using IoT technology. This solution provides a safe, efficient, and low-cost firefighting system while reducing human risk.

# PROPOSED METHOD

## OBJECTIVE:

The proposed system is a smart firefighting robot developed using the ESP32 microcontroller. It is designed to automatically detect fire and take immediate action to control it. Flame sensors and a  are used to identify the presence of fire.

Once fire is detected, a water pump is activated through a relay module to extinguish it. A servo motor helps in directing the water towards the fire source, improving effectiveness. The robot is made mobile using a motor driver so that it can move towards the affected area.

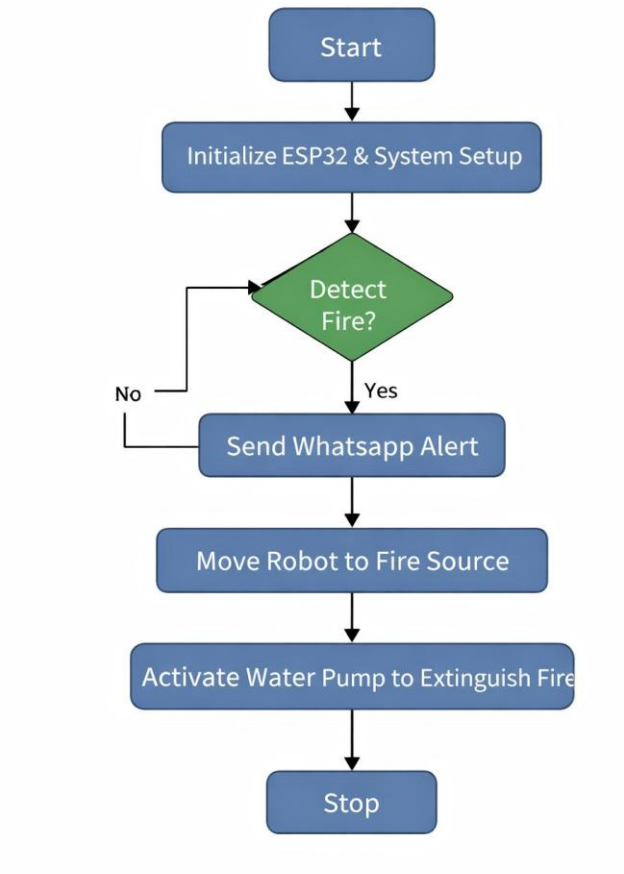
The system also includes IoT-based communication to send WhatsApp alerts to authorities, ensuring quick response. This method combines detection, action, mobility, and communication in a simple and efficient manner, making it a reliable and low-cost solution.

**Comparison of existing system and proposed system**

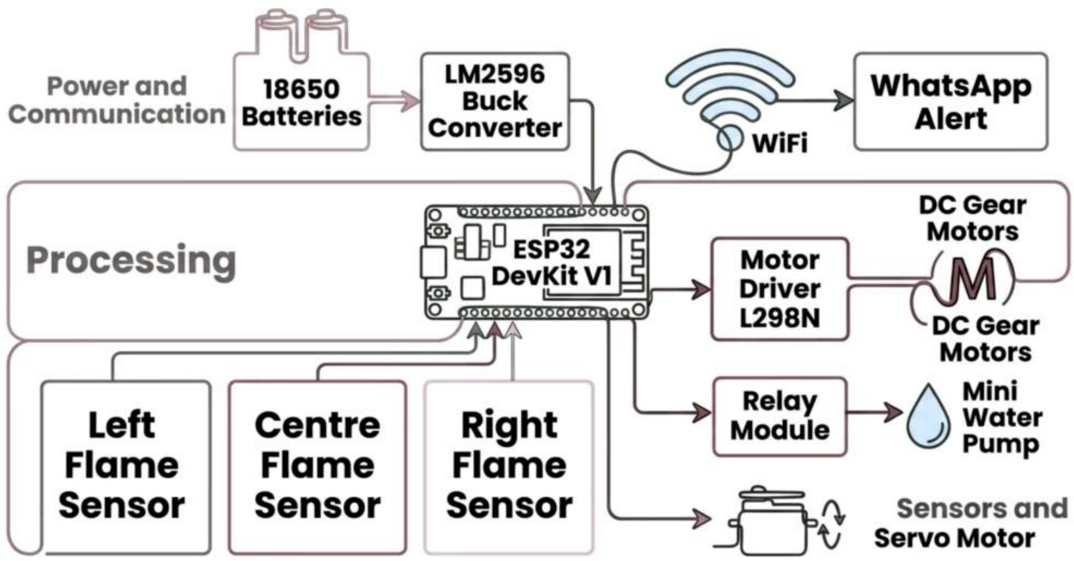
| FEATURE | EXISTING SYSTEM | PROPOSED SYSTEM |
|---|---|---|
| DETECTION | Basic sensors | Multiple sensors |
| EXTINGUISHING | Manual operation | Automatic system |
| MOBILITY | Fixed position | Mobile robot |
| ALERT SYSTEM | Slow | Immediate |
| EASE OF USE | Moderate | User friendly |
| COST | Medium | Lowcost |
| INNOVATION LEVEL | Less | High |

# METHODOLOGY

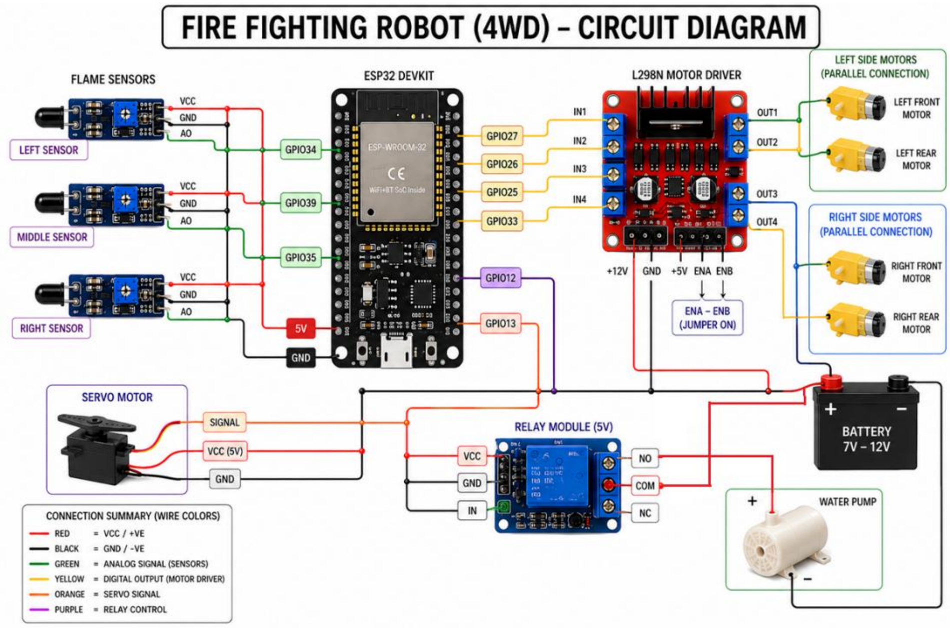
## FLOWCHART:




## BLOCK DIAGRAM:



# CIRCUIT DIAGRAM:




# COMPONENTS USED:

ESP32 DevKit V1 Board

Flame Sensor Module

SG90 Micro Servo Motor

Mini DC Water Pump (3–6V)

1-Channel Relay Module (5V)

Small Water Tank

Battery 3.7 V

Series Battery Holder (3 slot)

L298N 4wd motor driver

Breadboard

Jumper Wires

USB Cable for ESP32

## WORKING PRINCIPLE:

The firefighting robot operates based on sensor detection, control processing, and automatic action. The system is powered by a battery supply regulated through a buck converter to provide stable voltage to the ESP32 controller and other components.

Initially, the flame sensors continuously monitor the surroundings to detect the presence of fire. When a flame is detected, the sensor sends a signal to the ESP32 microcontroller, which processes the input.

Based on the sensor input, the ESP32 controls the movement of the robot using a motor driver, allowing it to move towards the direction of the fire. At the same time, a relay module is activated to switch on the water pump, which sprays water to extinguish the fire. A servo motor is used to adjust the direction of water flow for effective coverage.

In addition, the ESP32 uses its WiFi capability to send a WhatsApp alert to predefined users, informing them about the fire incident. This ensures quick communication and response.

Thus, the system performs fire detection, movement, fire suppression, and alert notification in an automatic and efficient manner, making it reliable for firefighting applications.

## SOFTWARE USED:

1. Arduino IDE
   - ✓ Used to write, compile, and upload code to ESP32
   - ✓ Supports C/C++ programming

✓ Provides Serial Monitor for debugging

2. ESP32 Board Package

✓ Installed inside Arduino IDE
✓ Enables programming of ESP32 microcontroller
✓ Provides libraries and board configurations

3. Twilio API

✓ Used to send WhatsApp alerts
✓ Works using HTTP requests
✓ Enables real-time notification system

4. Embedded C++

✓ Programming language used for development
✓ Controls sensors, motors, and communication

5. WiFi Library (ESP32)

✓ Used to connect ESP32 to internet
✓ Enables communication with cloud services

6. HTTPClient Library

✓ Used to send HTTP requests
✓ Helps in integrating Twilio API

7. ESP32Servo Library

✓ Used to control servo motor

**PROGRAM:**

```
#include <ESP32Servo.h>
#include <WiFi.h>
#include <HTTPClient.h>
#include <WiFiClientSecure.h>

// ---------------- PIN DEFINITIONS ----------------
#define IN1 27
#define IN2 26
#define IN3 25
#define IN4 33

#define pump 12
#define ServoPin 13

#define FlameLeft 34
#define FlameMiddle 39
#define FlameRight 35

#define LED 2

// ---------------- SETTINGS ----------------
int fireThreshold = 2000;
```

```
// ---------------- WIFI CONFIG ----------------
const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_PASSWORD";

// ---------------- TWILIO CONFIG ----------------
String account_sid = "YOUR_ACCOUNT_SID";
String auth_token  = "YOUR_AUTH_TOKEN";

String from = "whatsapp:+14155238886";
String to   = "whatsapp:+91XXXXXXXXXX";

 OBJECTS
Servo servoMotor;

 FLAGS
bool fireDetected = false;
bool alertSent = false;

 SETUP
void setup() {
  Serial.begin(115200);
```

```
pinMode(IN1, OUTPUT);
pinMode(IN2, OUTPUT);
pinMode(IN3, OUTPUT);
pinMode(IN4, OUTPUT);

pinMode(pump, OUTPUT);
pinMode(LED, OUTPUT);

pinMode(FlameLeft, INPUT);
pinMode(FlameMiddle, INPUT);
pinMode(FlameRight, INPUT);

servoMotor.attach(ServoPin);
servoMotor.write(90);

digitalWrite(pump, LOW);
digitalWrite(LED, LOW);

// WiFi Connection
WiFi.begin(ssid, password);
while (WiFi.status() != WL_CONNECTED) {
  delay(500);
}
```

```
}
LOOP
void loop() {

  int left = analogRead(FlameLeft);
  int middle = analogRead(FlameMiddle);
  int right = analogRead(FlameRight);

  fireDetected = (left < fireThreshold ||
                  middle < fireThreshold ||
                  right < fireThreshold);

  if (fireDetected) {

    digitalWrite(LED, HIGH);

    // Movement Logic
    if (middle < fireThreshold) {
      moveForward();
    }
    else if (left < fireThreshold) {
      turnLeft();
    }
```

```
else if (right < fireThreshold) {
  turnRight();
}

delay(300);
stopMoving();

// Pump Activation
digitalWrite(pump, HIGH);

// Servo Spray Motion
for (int angle = 60; angle <= 120; angle += 5) {
  servoMotor.write(angle);
  delay(100);
}
for (int angle = 120; angle >= 60; angle -= 5) {
  servoMotor.write(angle);
  delay(100);
}

digitalWrite(pump, LOW);

// Send WhatsApp Alert (only once)
```

```
    if (!alertSent) {
      sendAlert("Fire Detected! Robot Activated.");
      alertSent = true;
    }

  } else {
    stopMoving();
    digitalWrite(pump, LOW);
    digitalWrite(LED, LOW);
    servoMotor.write(90);
    alertSent = false;
  }

  delay(300);
}
 MOTOR FUNCTIONS
void moveForward() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}
```

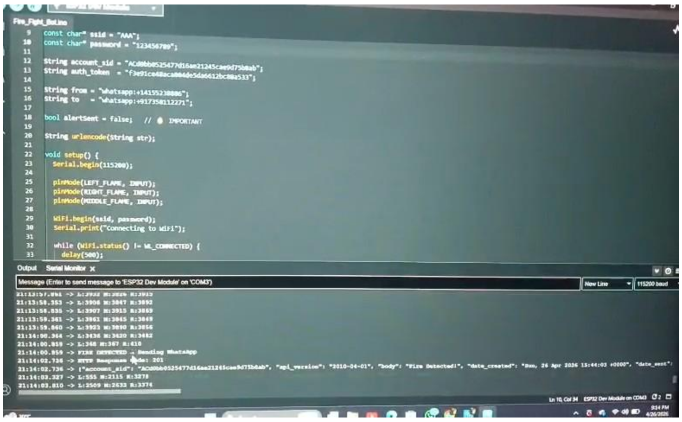
```
void turnLeft() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

void turnRight() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void stopMoving() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
}
```

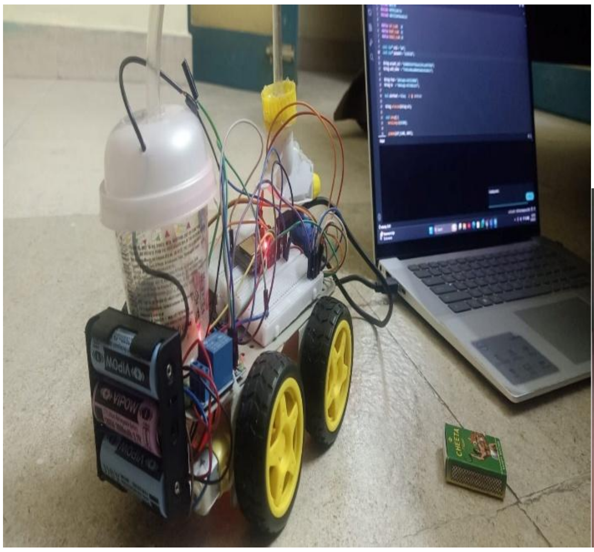
WHATSAPP ALERT

```
void sendAlert(String message) {
```

```
  if (WiFi.status() == WL_CONNECTED) {

    WiFiClientSecure client;
    client.setInsecure();

    HTTPClient http;

    String url = "https://api.twilio.com/2010-04-01/Accounts/"
                 + account_sid + "/Messages.json";

    http.begin(client, url);
    http.setAuthorization(account_sid.c_str(),
auth_token.c_str());
    http.addHeader("Content-Type", "application/x-www-form-
urlencoded");

    String body =
"From=whatsapp:%2B14155238886&To=whatsapp:%2B91X
XXXXXXXXX&Body=" + message;

    http.POST(body);
    http.end();
  }
}
```

# RESULT AND DISCUSSION:



## OBSERVATIONS:

Figure 1: WhatsApp Alert Output

The system successfully sends real-time alerts when fire is detected. The notification message “Fire Detected!” is received instantly on the mobile device through WhatsApp. This confirms that the ESP32 WiFi communication is working properly and ensures quick information delivery during emergency situations. The repeated alerts indicate continuous detection of fire until the condition is cleared.

Figure 2: Firefighting Robot Prototype

The developed firefighting robot prototype operates effectively by detecting fire and responding automatically. The robot is able to move towards the fire source using DC motors controlled by the motor driver. Once fire is detected, the water pump is activated through the relay module to extinguish the fire. The hardware components are properly integrated, and the system performs detection, movement, and fire suppression efficiently.

## PROTOTYPE DEVELOPMENT AND TEAM PHOTO:

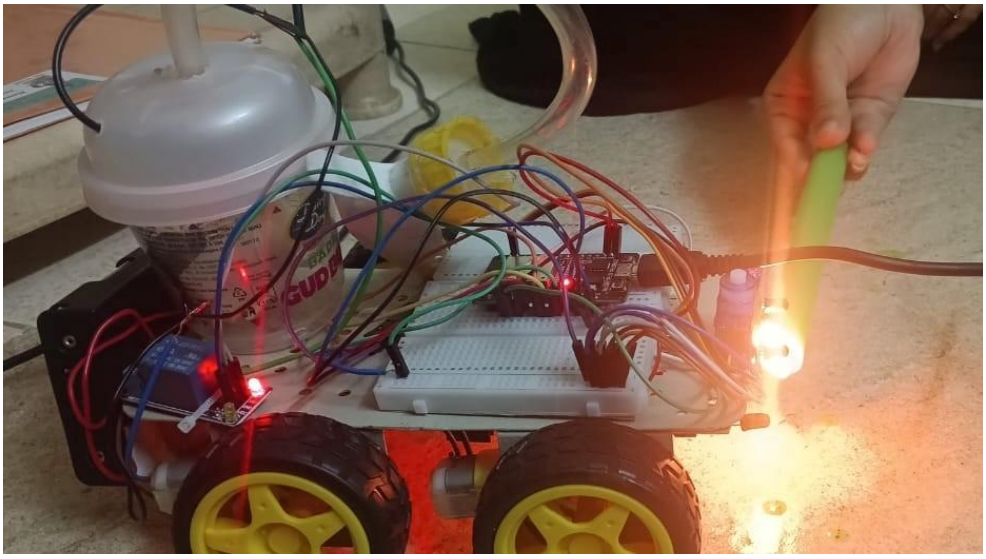



**Figure 1:Fire Fighting Robot Prototype**

**Figure 2:Team with Prototype**

# CONCLUSION AND FUTURE WORK

## CONCLUSION:

This project successfully presents the design and development of a firefighting robot using ESP32 with a WhatsApp alert system. The system is capable of detecting fire using sensors and automatically taking action to extinguish it using a water pump. The integration of mobility enables the robot to move towards the fire source, improving its efficiency.

The major benefit of this system is the reduction of human risk, as it minimizes the need for direct human involvement in dangerous situations. The use of IoT technology enhances the system by providing real-time communication through WhatsApp alerts, ensuring quick response during emergencies. The system is cost-effective, reliable, and easy to implement.

This project can be applied in various real-time environments such as homes, industries, laboratories, and warehouses, where early fire detection and control are essential.

## FUTURE WORK:

- ✓ Integration of camera for live monitoring
- ✓ Implementation of AI for smart navigation
- ✓ Addition of advanced fire and gas sensors
- ✓ Improvement in battery life and power efficiency
- ✓ Expansion for large-scale industrial applications

# EXPENDITURE DETAILS :

| S.NO | NAME | MODEL NO | SPECIFICATION | AMOUNT |
|---|---|---|---|---|
| 1 | **ESP32 BOARD** | **WIFI** | **MCU** | **350** |
| 2 | **FLAME SENSOR** | **KY-026** | **IR** | **150** |
| 3 | **SERVO MOTOR** | **SG90** | **180DEGREE ROTATION** | **160** |
| 4 | **MINI WATER PUMP** | **3-6V** | **SUBMERSIBLE PUMP** | **180** |
| 5 | **MOTOR DRIVER** | **L298N** | **DUAL BRIDGE** | **200** |
| 6 | **BATTERIES** | **3 SLOT** | **SERIES HOLDER** | **100** |
| 7 | **BREADBOARD** | | **MINI GENERAL** | **100** |
| 8 | **JUMPER WIRES** | | **MALE-MALE/FEMALE-FEMALE** | **100** |
| 9 | **RELAY MODULE** | **5V** | **SINGLE CHANNEL** | **150** |
| | | | **TOTAL** | **1490** |

# ROLES AND RESPONSIBILITIES:

| S.NO | TEAM MEMBERS | REG NO | SOFTWARE | HARDWARE |
|---|---|---|---|---|
| 1 | ANUSRI | **113323106004** | **WHATSAPP INTEGRATION** | **MOTOR DRIVER SETUP** |
| 2 | DANISHTAA | **113323106018** | **DOCUMENTATION** | **SENSOR SETUP** |
| 3 | SHARMILA | **113323106094** | **CODE TESTING** | **PUMP SETUP** |
| 4 | SRIDHARSHINI | **113323106097** | **IOT SETUP** | **CIRCUIT DESIGN** |
| 5 | THIRILOSHANI | **113323106108** | **PROGRAMMING** | **FINAL ASSEMBLY** |
| 6 | VAISHNAVI | **113323106111** | **REPORT PREPARATION** | **WIRING CONNECTIONS** |

## REFERENCES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"Arduino IDE Software," Arduino IDE, Available: https://www.arduino.cc□

## ACHIEVED Pos and PSOs:

| S.NO | POs | PO Statement |
|---|---|---|
| 1 | PO1 | Apply engineering knowledge to design and develop the firefighting robot system. |
| 2 | PO2 | Identify and analyze real-world problems related to fire safety and automation. |
| 3 | PO3 | Design a solution using ESP32 and IoT for effective fire detection and control. |
| 4 | PO4 | Conduct experiments and interpret results for system performance. |
| 5 | PO5 | Use modern tools like Arduino IDE and IoT platforms for development. |

| S.NO | PSOs | PSO Statement |
|---|---|---|
| 1 | PSO1 | Apply embedded system concepts to develop IoT-based applications. |
| 2 | PSO2 | Design and implement hardware circuits using sensors and microcontrollers. |
| 3 | PSO3 | Develop automation systems for real-time monitoring and control. |